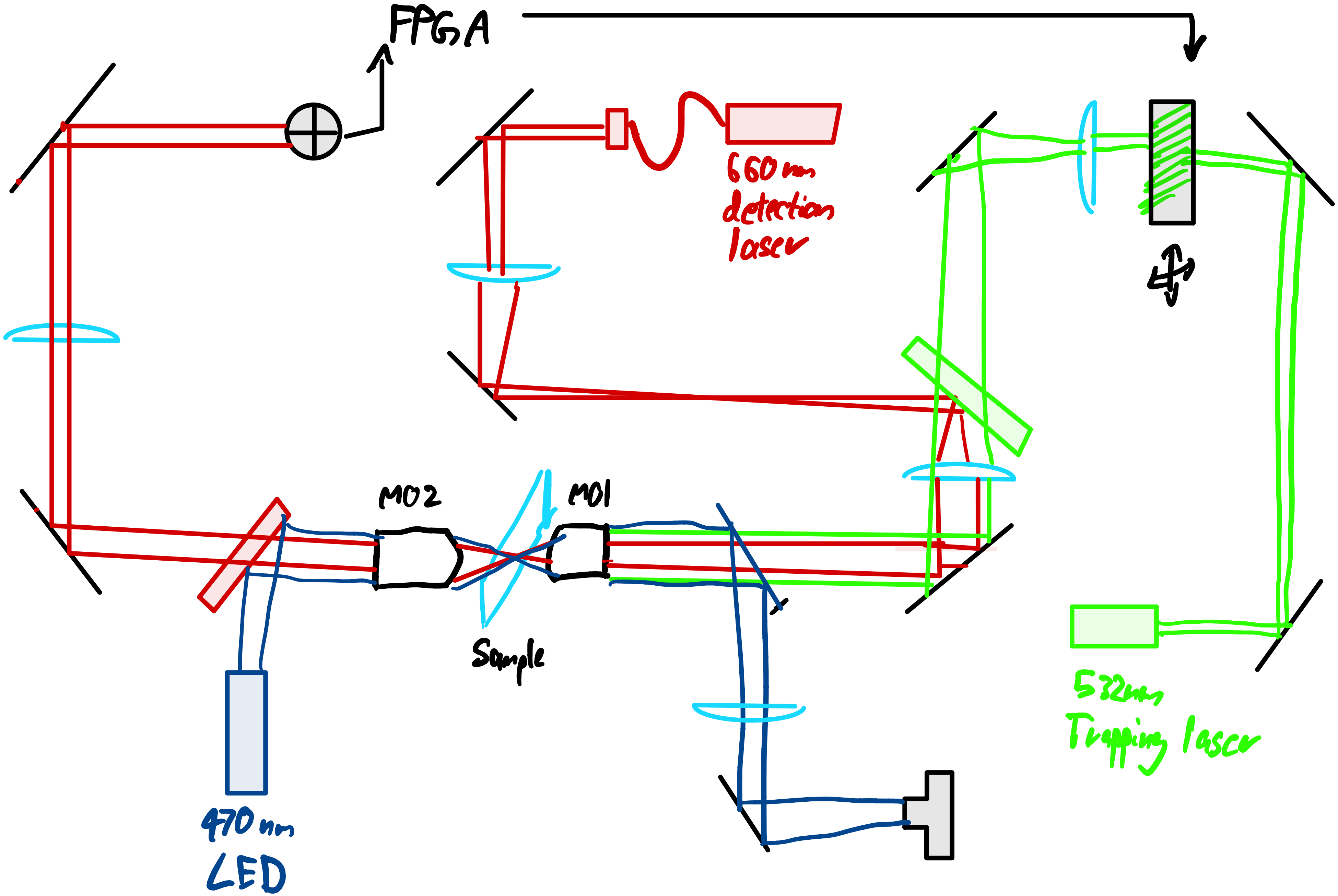
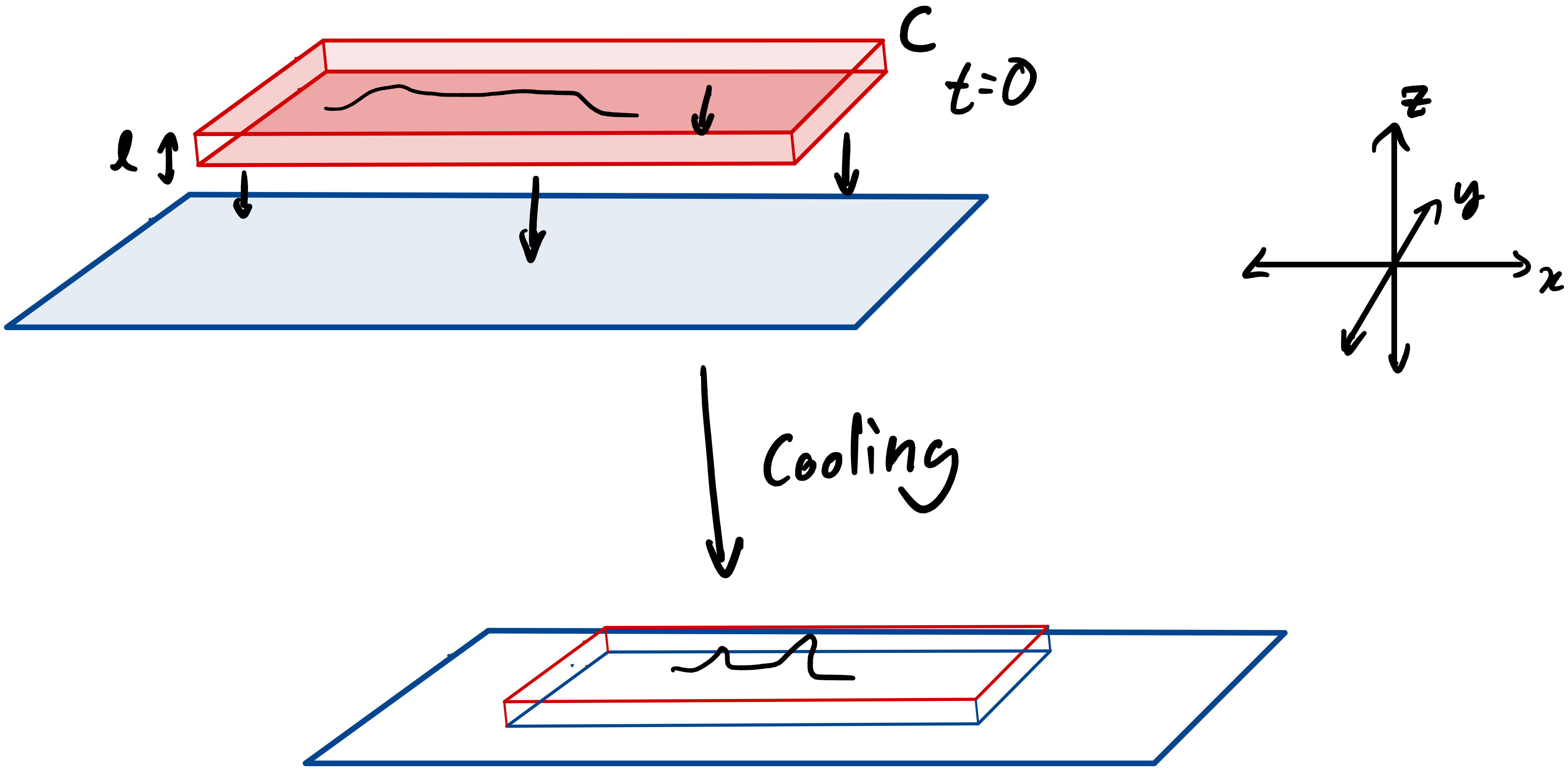
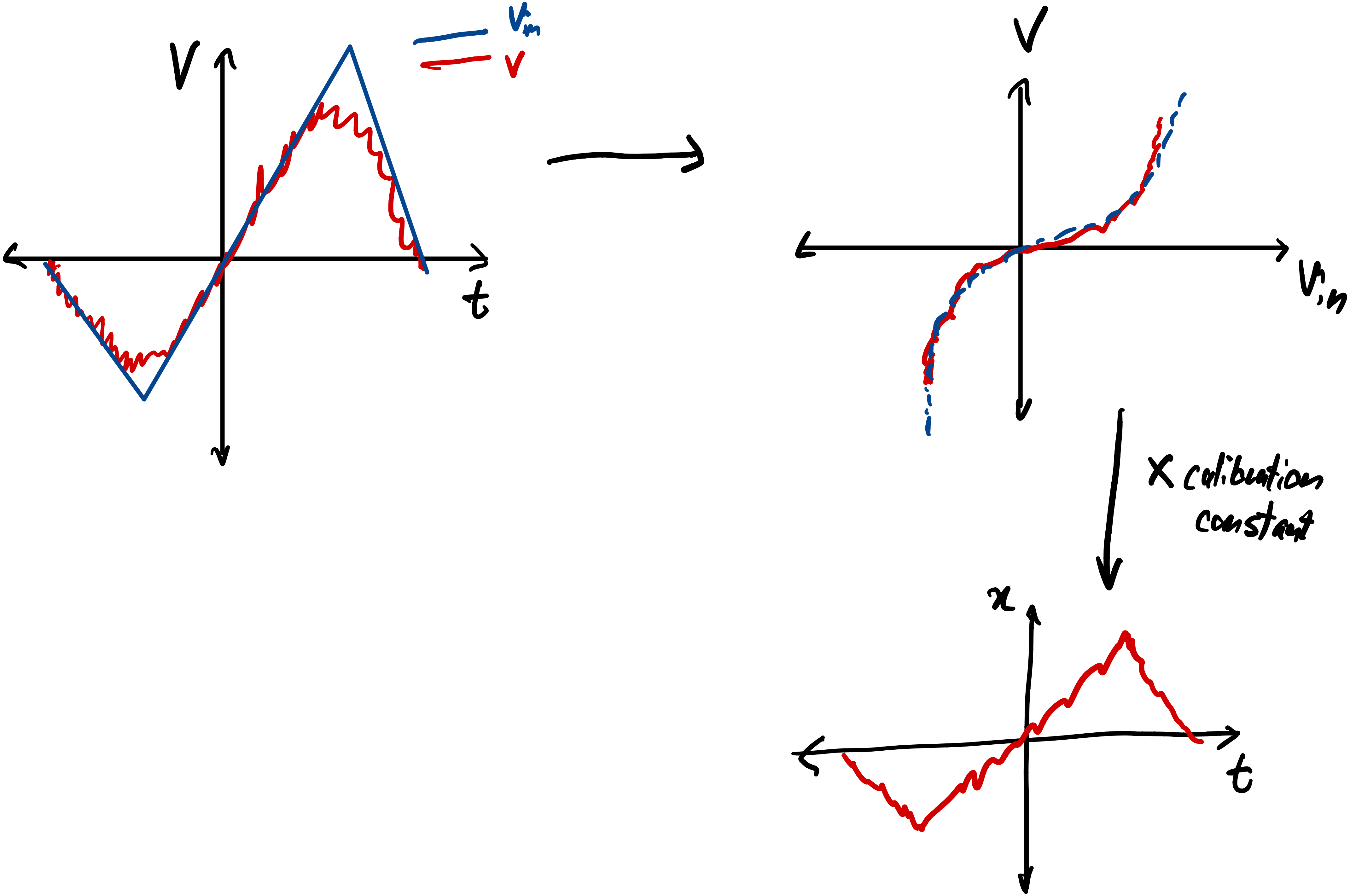
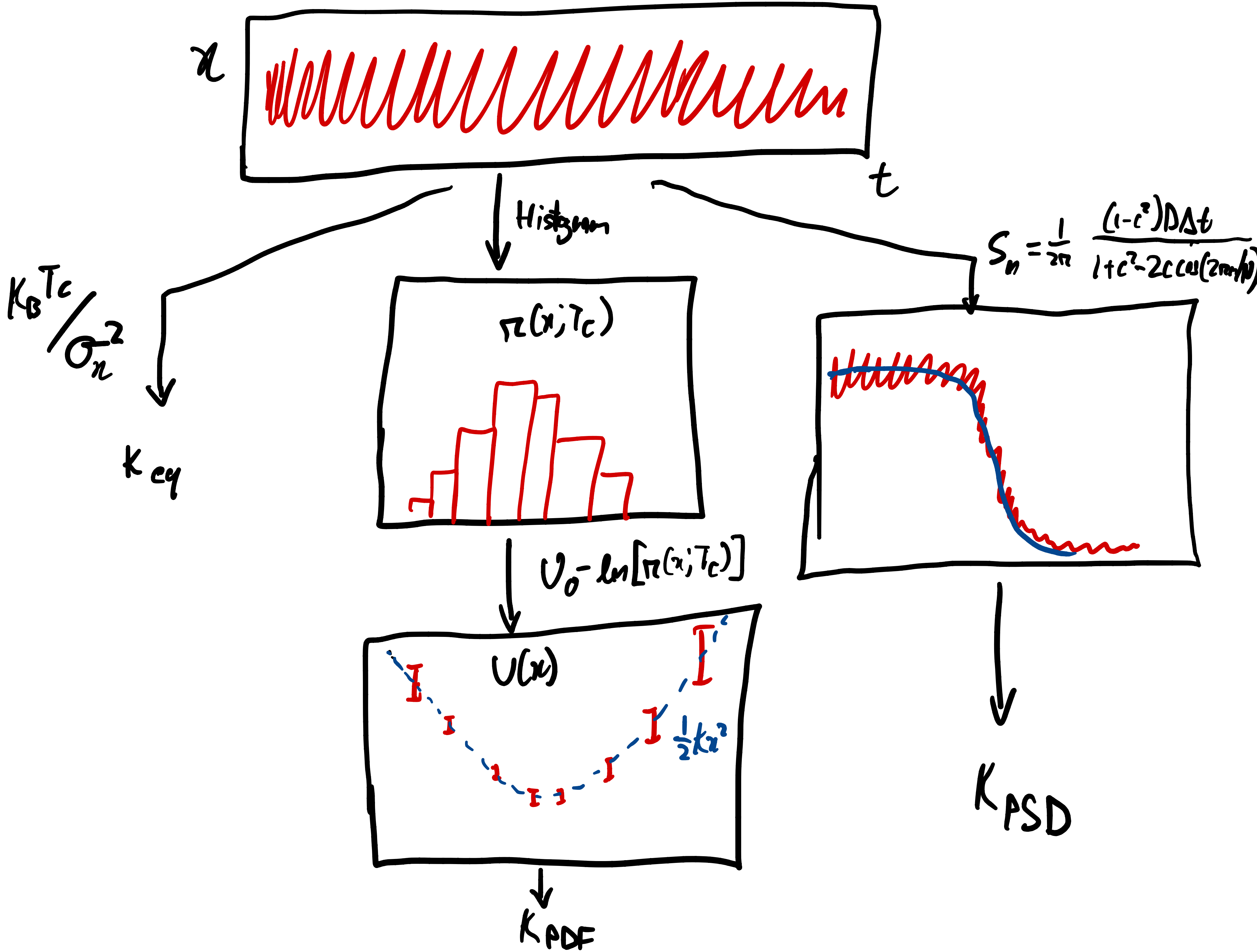
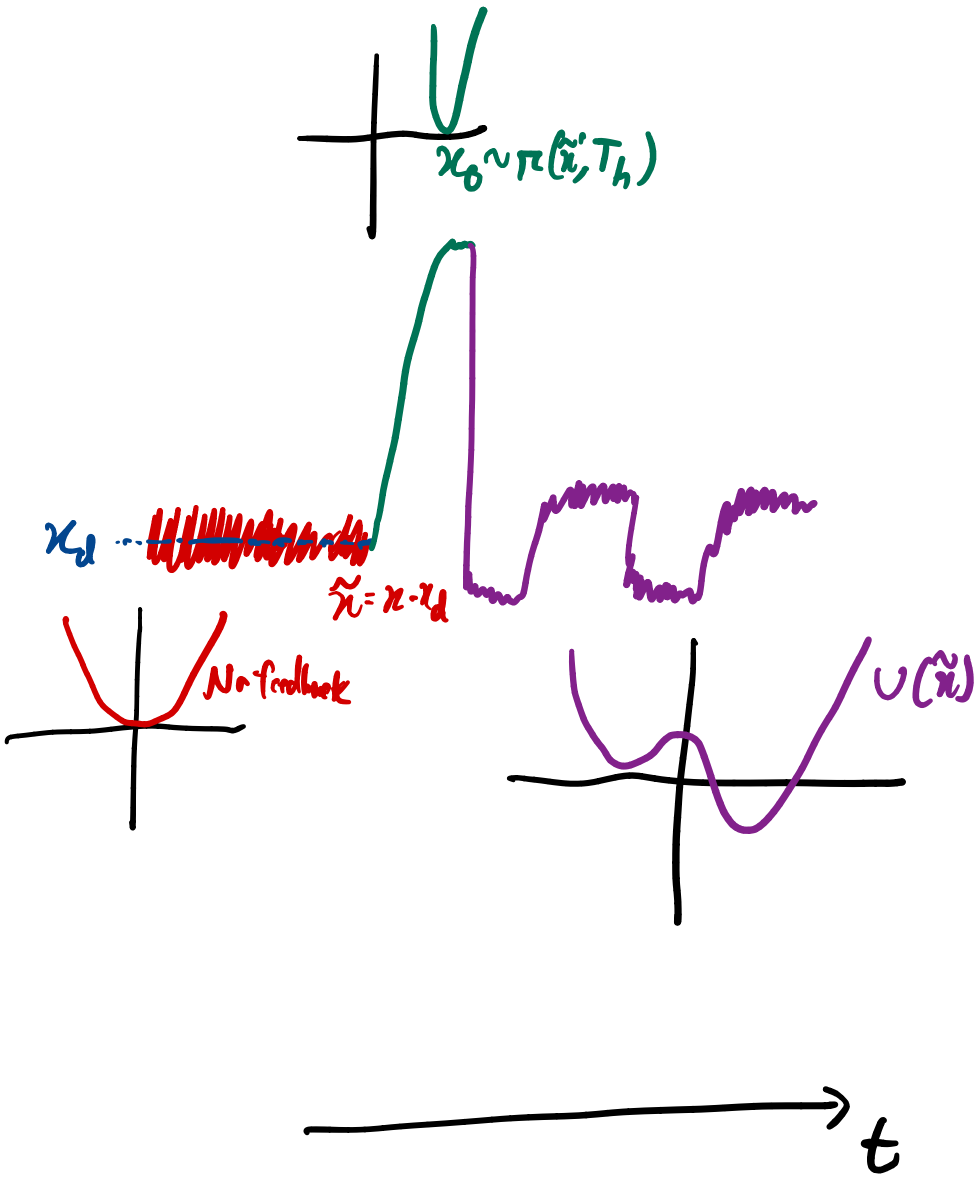

FPGA
660 nm
detection
laser
MO2
MO1
Sample
470 nm
LED
532 nm
Trapping laser


C
t=0
l
Cooling
z
y
x

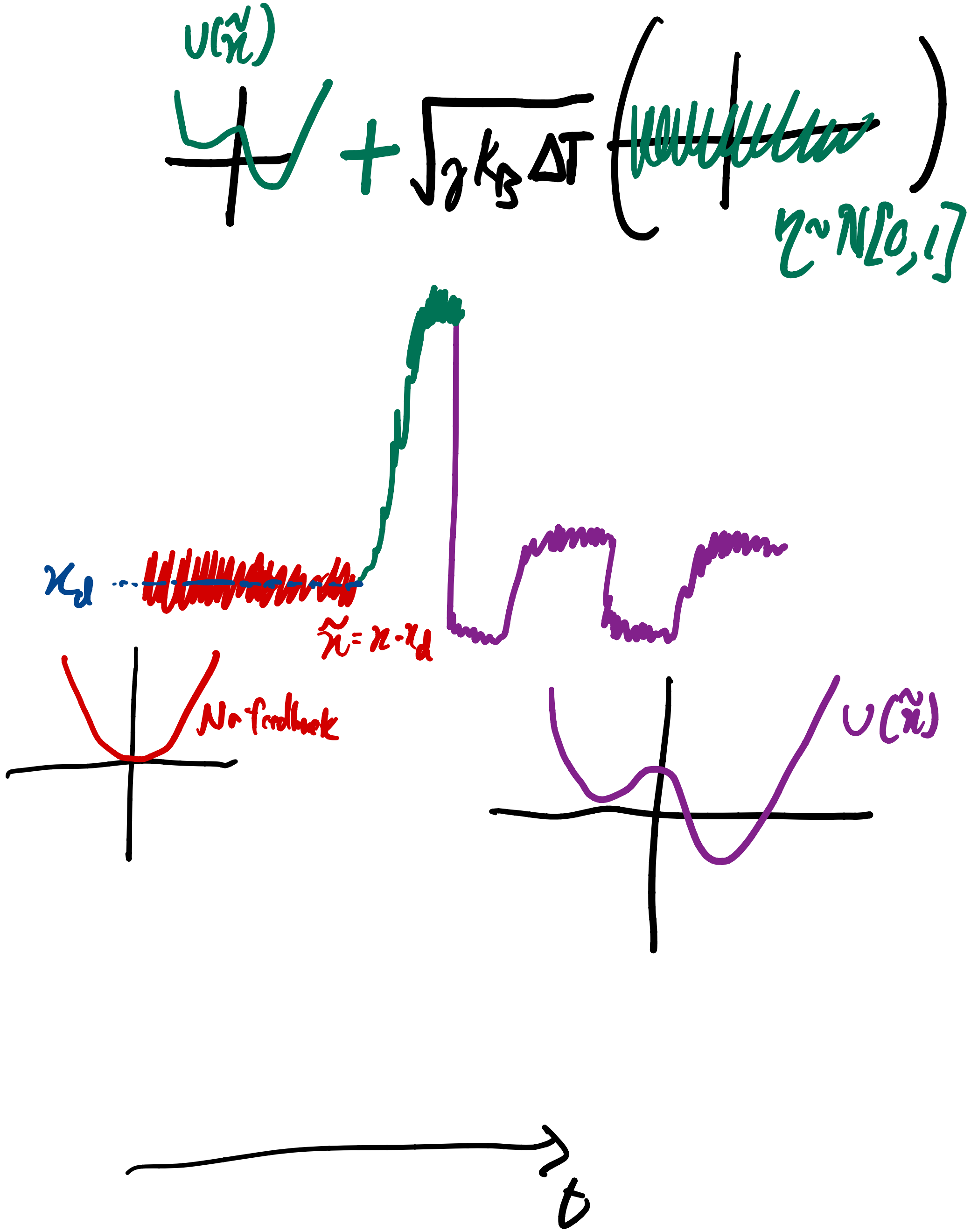


V
t
V_in
V
V
V_in
× calibration constant
x
t

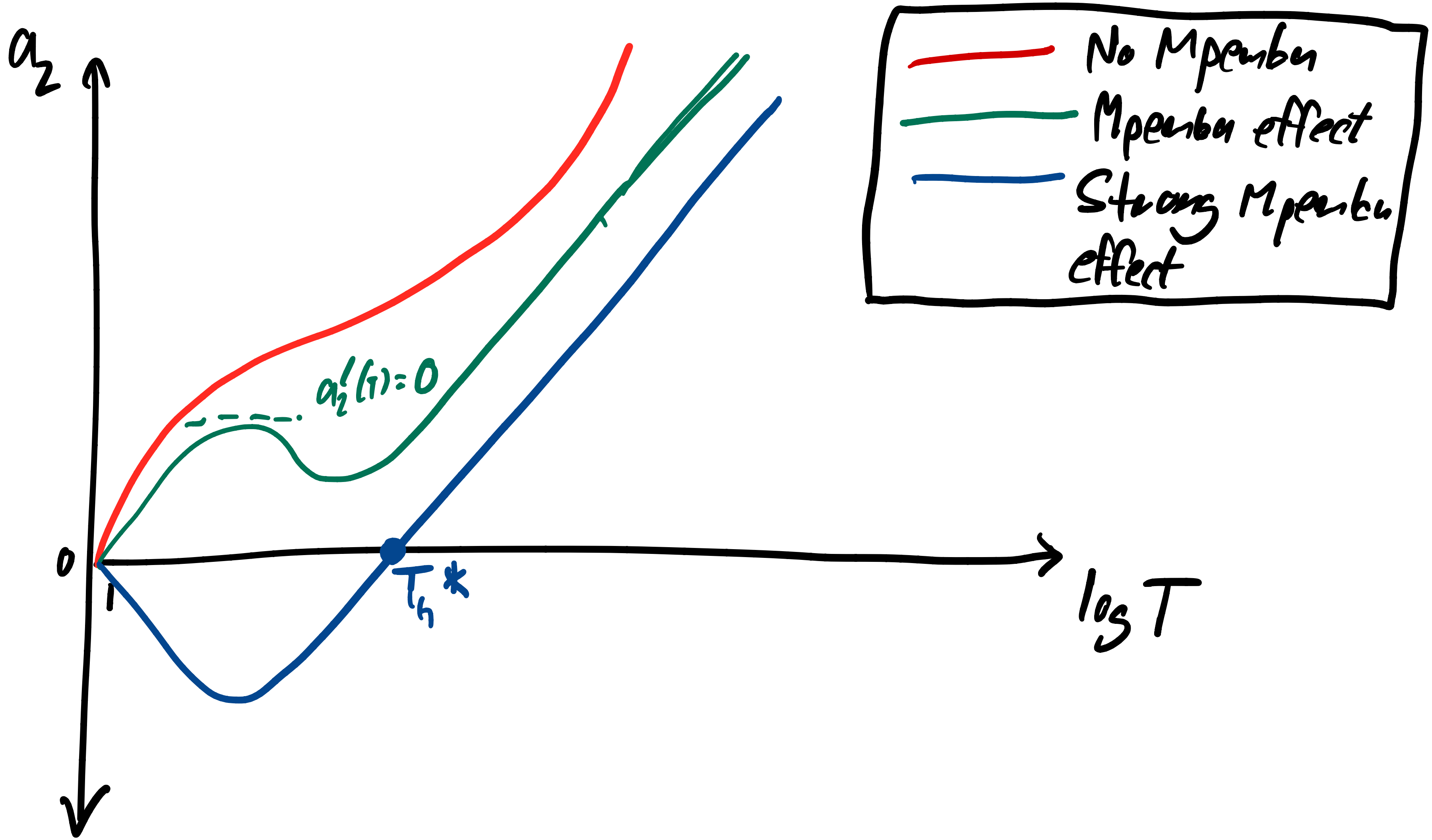

x
t
Histogram
$\pi(x;T_c)$
$U_0 - \ln[\pi(x;T_c)]$
$U(x)$
$\frac{1}{2}kx^2$
K_{PDF}
$K_B T_c / \sigma_x^2$
K_{eq}
$S_n = \frac{1}{2\pi} \frac{(1-c^2)D\Delta t}{1+c^2-2c\cos(2\pi n/N)}$
K_{PSD}



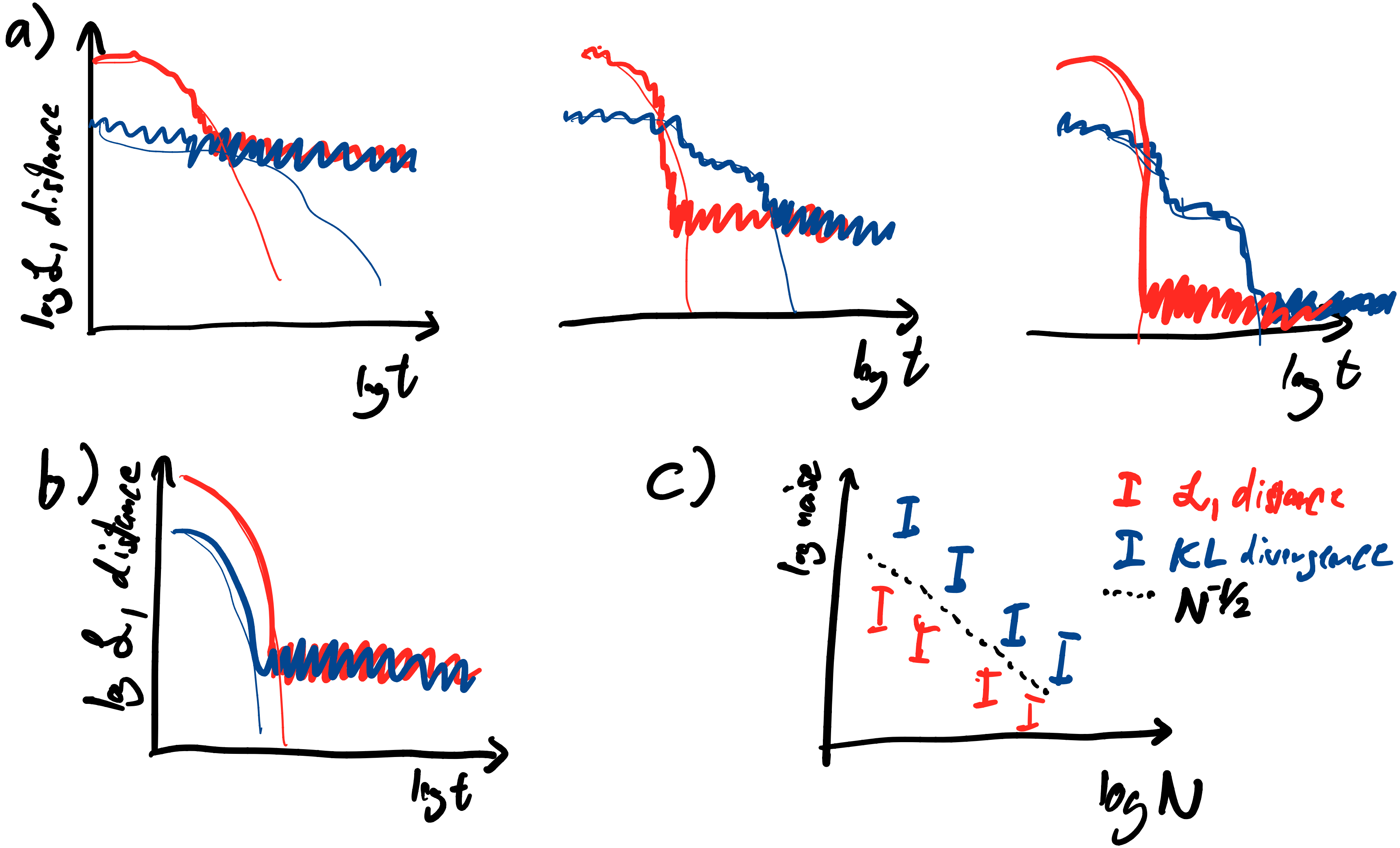
$x_0 \sim \pi(\tilde{x}; T_h)$
x_d
$\tilde{x} = x - x_d$
No feedback
$U(\tilde{x})$
t



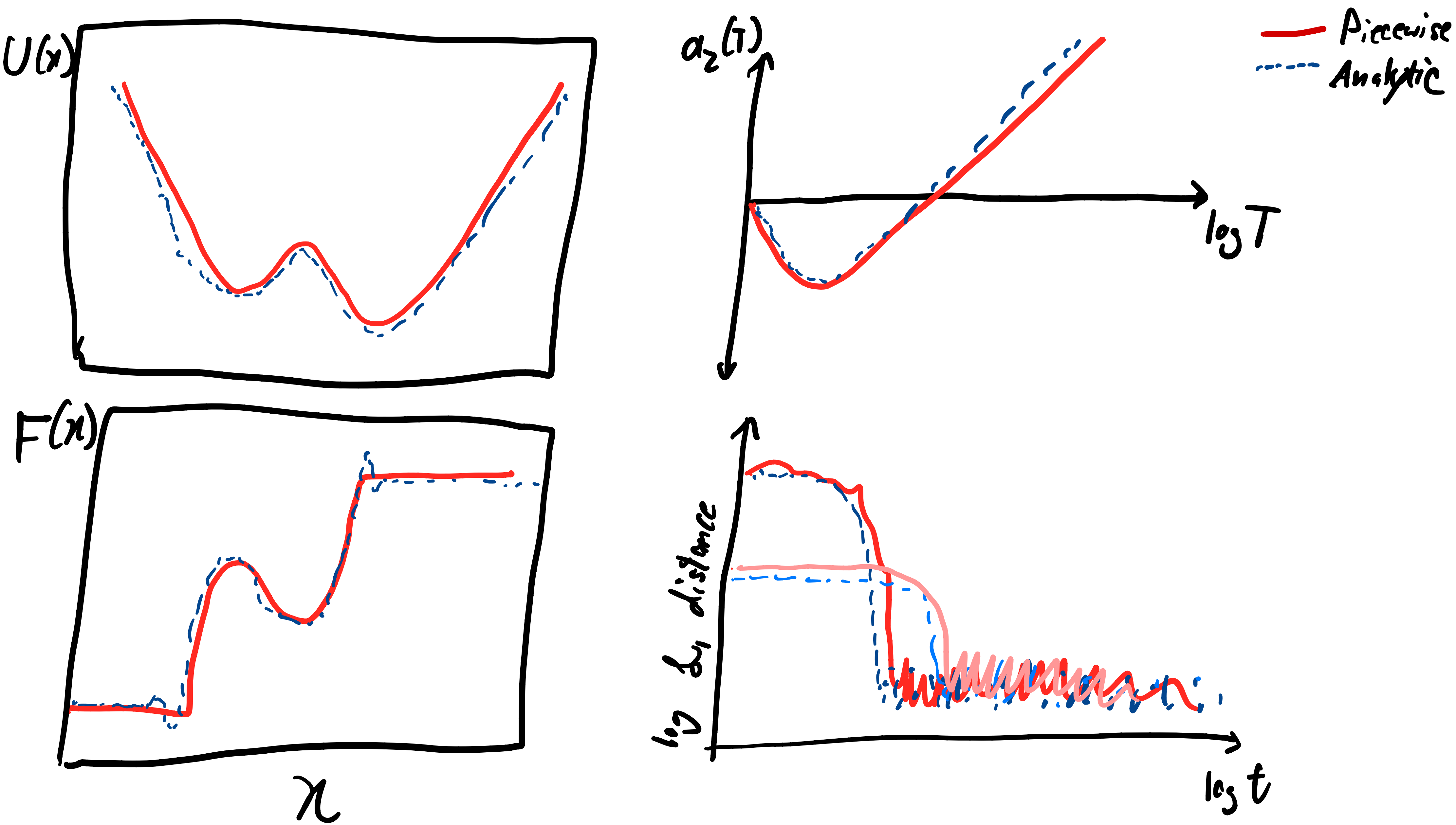
$U(\tilde{x})$
$+\sqrt{\gamma k_B \Delta T}$
$\eta \sim N[0,1]$
x_d
$\tilde{x} = x - x_d$
No feedback
$U(\tilde{x})$
t

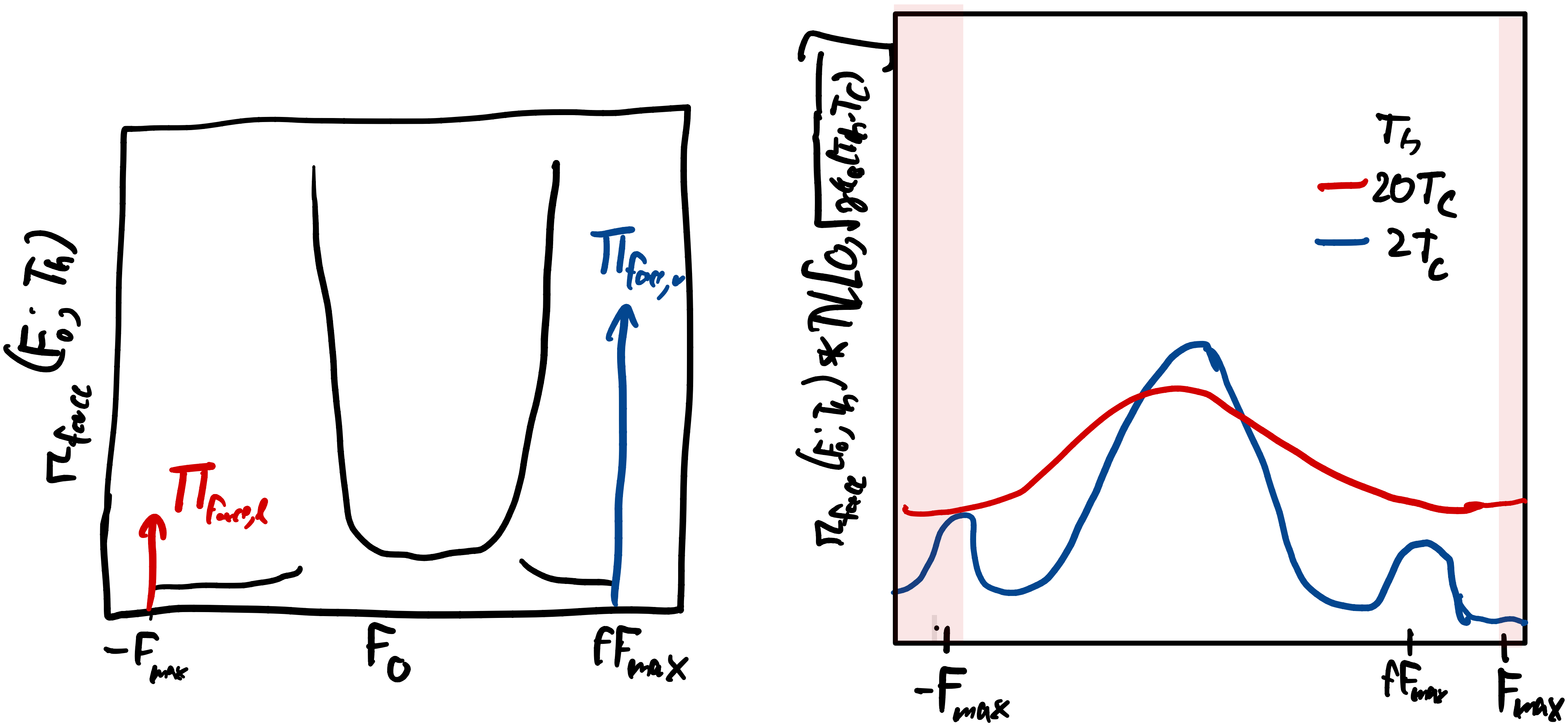


a_2
0
1
$a_2'(T)=0$
T_h^*
$\log T$
No Mpemba
Mpemba effect
Strong Mpemba effect



a)
log $\mathcal{L}_1$ distance
log t
log t
log t
b)
log $\mathcal{L}_1$ distance
log t
c)
log noise
log N
$\mathcal{L}_1$ distance
KL divergence
$N^{-1/2}$


U(x)
F(x)
x
a_2(T)
log T
log L_1 distance
log t
Piecewise
Analytic



$\pi_{force}(F_0; T_h)$
$\Pi_{force,d}$
$\Pi_{force,u}$
$-F_{max}$
F_0
fF_{max}
$\pi_{force}(F_0; T_h) * \mathcal{N}[0, \sqrt{g\ell_0(T_h - T_C)}]$
T_h
$20T_C$
$2T_C$
$-F_{max}$
fF_{max}
F_{max}